

Nucleus Structure & Cell Cycle - Factoid Sheet

Week 4 - Monday - Core Lecture 2 - current faculty deck + official Week 4 objectives

Master chain: nuclear structure → gated transport → lamin integrity → cyclin/CDK control → Rb/E2F G1-S restriction point.

Nucleus Structures

Structure	Function / anchor
Nuclear envelope	Compartment boundary; outer membrane continuous with ER
Nuclear pore complex	Selective cytoplasm-nucleus exchange
Nucleolus	rRNA synthesis/processing + ribosomal-subunit assembly
Nuclear matrix/scaffold	Nuclear architecture, chromosome territories, transient DNA attachment
Nuclear lamina	Lamins A/B/C; mechanical support + chromatin organization
Euchromatin	Open / transcriptionally accessible
Heterochromatin	Dense / less active

Nuclear Transport

Small cargo
Roughly <40 kDa can cross more freely by diffusion.

Large cargo
NLS/NES + karyopherin (importin/exportin) + Ran-GTP cycle.

Import sequence: NLS cargo binds importin alpha/beta → pore transit → Ran-GTP promotes cargo release → receptors recycle; RanGAP converts Ran-GTP to Ran-GDP in cytoplasm.

Import / Export Logic

Step	Import anchor
1	NLS cargo binds importin alpha; importin beta joins
2	Complex crosses NPC
3	Nuclear Ran-GTP binding promotes cargo release
4	Import receptors recycle to cytoplasm
5	RanGAP drives Ran-GTP → Ran-GDP in cytoplasm

Nucleolus / Matrix

Nucleolus
rRNA transcription/processing; 18S, 5.8S, 28S; ribosomal-subunit assembly; multiple nucleoli can reflect active growth.

Matrix/scaffold
Dynamic chromosome territories; architecture; transient DNA attachment; replication/transcription/repair organization.

Lamina / Laminopathies

Anchor	Fact
Lamins	10-12 nm intermediate-filament proteins
LMNA	Lamins A and C from same gene by alternative splicing
HGPS	Mutant lamin A product = progerin; abnormal nuclear architecture
Other examples	Cardiomyopathy; congenital muscular dystrophy
Nuclear envelopathy	Can involve interacting envelope proteins such as emerin
Zokinvy	Farnesylation inhibitor treatment anchor in deck

Mitosis Trigger

MPF = cyclin B + CDK1. Drives chromosome condensation, lamin phosphorylation/nuclear-envelope breakdown, spindle formation, and organelle fragmentation.

Nucleus / Cell Cycle - Fast Recall

Major Cyclin-CDK Pairs

Phase	Complex	Anchor
G1	Cyclin D - CDK4/6	Growth factor response; Rb phosphorylation
G1/S	Cyclin E - CDK2	S-phase commitment
S	Cyclin A - CDK2	DNA replication
G2	Cyclin A - CDK1	Prepare for mitosis
M	Cyclin B - CDK1	MPF

Cell-Cycle Phase Framework

G1 Growth + restriction point.	S DNA replication.	G2/M Prepare for and execute mitosis.
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How Activity Is Controlled

Cyclins Concentration changes; ubiquitin-proteasome degradation.	CDKs Cyclin binding + activating/inhibitory phosphorylation; CKIs such as p21 block activity.
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Rb Restriction Point

Unphosphorylated Rb = active brake. It sequesters E2F. Growth-factor signaling activates cyclin D/CDK4/6 → Rb phosphorylation → Rb inactive → E2F released → S-phase genes expressed.
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CDK4/6 Inhibition

Block CDK4/6 → less Rb phosphorylation → Rb remains active → E2F remains restrained → G1-S progression slows/stops.

Growth Factor to S Phase

Growth factor → receptor / signal-transduction cascade (e.g., Ras-pathway kinase signaling) → increased cyclin D/CDK4/6 activity → Rb phosphorylation/inactivation → E2F release → S-phase transcription.

Cell-State Anchors

State	Recall
G0	Quiescent, metabolically active, not proliferating
Actively cycling epithelium	Frequent division
Hepatocyte	Low-frequency division; can re-enter cycle
Neuron	Generally non-dividing differentiated state

Mitotic CDK1/Cyclin B Outputs

Target/event	Outcome
Histone H1 phosphorylation	Chromatin condensation
Lamin phosphorylation	Lamina depolymerization + nuclear-envelope breakdown
Spindle machinery	Mitotic spindle formation
Golgi/ER/mitochondrial organization	Fragmentation/remodeling during mitosis

Must-Not-Miss Traps

Lamins = intermediate filaments.	Cyclin B/CDK1 = MPF.
Phosphorylated Rb = inactive.	CDK4/6 inhibitors preserve the Rb brake.

30-Second Review

NLS/importins/Ran for large nuclear import. Nucleolus builds ribosomal components. LMNA disease can create progerin. Cyclins rise/fall; CDKs are switched by binding/phosphorylation. D-CDK4/6 phosphorylates Rb; E2F is released; S phase follows.

Assigned ScholarRx

Nucleus: <https://exchange.scholarrx.com/brick/nucleus> | Cell Cycle: <https://exchange.scholarrx.com/brick/cell-cycle>